Lesson plan · for the teacher

# Start from a problem

IoT · Level 4 · Lesson 1

**Time**
40 min

**Materials**
The board and chalk, or large sheets of paper and markers; A one-page plan template: six boxes drawn on the board; Sticky notes or slips of paper for voting; Optional: last term's record of the problem (for example, dry days from the caretaker)

## Learning goals

- Turn a school problem into a measurable question and a success measure.
- Choose sensors, power and network by weighing needs, cost and upkeep with the people affected.
- Write a one-page project plan with roles for pupils.

## Activity: Problem hunt

1. Warm-up (5 min): Tell the story of the box of smart sensors that sat in a cupboard for a year. Ask: why did nobody use them? Collect answers on the board.
2. Teach (8 min): Work one example with the class: the water tank. Turn it into a question with a quantity, a place and a time, then a success measure with a number, a date and a baseline.
3. Hunt (10 min): In groups of four, pupils list three problems at school. For each, they write who it hurts, what could be measured, and how they would know it got better.
4. Vote (5 min): Each group reads out its best problem in one sentence. The class votes with a show of hands or sticky notes.
5. Plan (12 min): Draw six boxes on the board: problem, question, success measure, parts and costs, timeline with a pilot, pupil roles. Fill them in together for the winning problem.

**Before class, ask the caretaker or a cleaner which school problem they would fix first: their answer makes a good example. Keep the hunt to problems, not complaints about people, and use no names.**

## Discussion

- Which problem did your pupils choose, and who does it hurt most?
- What could you measure first with no sensor at all?
- Who outside the class must agree before you build anything?

## Exit ticket

- Write one measurable question and one success measure for a school problem.

**Try it at school tomorrow**
Problem hunt: With your class, list three problems at school. For each, pupils write who it hurts, what could be measured, and how they would know it got better. The class votes for one and drafts its one-page plan together: question, success measure, parts and costs, a pilot, and roles for pupils.

Practice questions · for you

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**1** **Your pupils scored three designs for the tank sensor, and LoRa came out on top. Nobody has asked the caretaker, who will change the batteries and climb the tank. What should happen next?**

- ◯ Drop the matrix and let the caretaker choose alone
- ◯ Agree the weights with him, then score again
- ◯ Order the LoRa parts: the scores have decided

**2** **Which of these are measurable questions a sensor project could answer?**

- ☐ Do pupils care about saving water?
- ☐ On how many holiday days does the garden soil drop below 20% moisture?
- ☐ At what time does Room 4 first go above 1,500 ppm of $CO_2$?
- ☐ How fast does the tank level fall on school days?
- ☐ How can we make our school smarter?

**3** **Score two networks for the tank sensor. The caretaker weighted cost 2 and upkeep 5. Wi-Fi scores 4 for cost and 2 for upkeep; LoRa scores 3 for cost and 5 for upkeep.**

Wi-Fi: 2 × 4 + 5 × 2 = _____. LoRa: 2 × 3 + 5 × 5 = _____. With these weights, _____ comes out on top.

**Word bank:** 6 · Wi-Fi · 18 · LoRa · 31

**4** **Put the start of a school sensor project in a sensible order.**

- ☐ Turn the problem into a measurable question
- ☐ Measure by hand to get a baseline
- ☐ Write the one-page plan, starting with a pilot
- ☐ Talk to the people the problem affects
- ☐ Weigh the options with the people affected

**5** **Match each part of the one-page plan to its example from the water-tank project.**

| | | |
|---|---|---|
| Measurable question | • • • • | Five dry days last term |
| Success measure | • • • • | How fast does the level fall on school days? |
| Baseline | • • • • | A SIM data plan, paid every month |
| Running cost | • • • • | The data team checks the graph each week |
| Pupil role | • • • • | No dry days by the end of term |

✎ **Which problem at your school do people feel most? Whom should you talk to before choosing anything?**

Check your answers

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**1** **Agree the weights with him, then score again**

The weights are a judgement, so the people who live with the system should help set them; the caretaker may care most about upkeep.

**2** **How fast does the tank level fall on school days?; On how many holiday days does the garden soil drop below 20% moisture?; At what time does Room 4 first go above 1,500 ppm of $CO_2$?**

Each right one names a quantity, a place and a time, so readings can answer it.

**3** **Wi-Fi: 2 × 4 + 5 × 2 = 18. LoRa: 2 × 3 + 5 × 5 = 31. With these weights, LoRa comes out on top.**

Multiply each score by its weight and add: Wi-Fi 8 + 10 = 18, LoRa 6 + 25 = 31.

**4** **Top to bottom: 2, 3, 5, 1, 4**

Each step needs the one before: the people affected shape the question; you can't measure without a question; measuring by hand gives the baseline and may show that no sensor is needed (a tap left running), so it comes before choosing parts; and the plan comes last because it records what you chose.

**5** **Measurable question — How fast does the level fall on school days?**
**Success measure — No dry days by the end of term**
**Baseline — Five dry days last term**
**Running cost — A SIM data plan, paid every month**
**Pupil role — The data team checks the graph each week**

The question says what to measure, the success measure says what "worked" means, and the baseline is where you start from.